

Credit risk

Purpose: Modelling the event of default.
Define T_d , the time of default, symbolically as:
$$T_d = \inf \{t \geq 0 \mid \underbrace{V(t)}_{\text{Value of firm}} < \underbrace{B(t)}_{\text{Value of debt}}\}$$

Problem: $V(t)$ and $B(t)$ are not observable.
Fix: We want to calculate the conditional default/survival probability:
 $p_d(t,T) = P(T_d \leq T | \mathcal{F}_t)$, $1 - p_d(t,T) = P(T_d > T | \mathcal{F}_t)$, $\forall t \in [0,T]$
 $p_d(t,T) = P(T_d \leq T | \mathcal{F}_t)$
Tractable model $\rightarrow$ explicit formula for p_d .
Structural model $\rightarrow$ no explicit formula.

- Simple Assumptions:
- $T_d > 0$ is a random variable.
 - Define $H(t) = 1_{\{T_d \leq t\}}$, $H_t = \sigma(H(s) | s \leq t)$
 - H_t : Information on whether default has happened at time t (or before)
 - G_t : The remaining information in the market. Price history, stock, vol, macrodata.
 - $\mathcal{F}_t$: The market filtration $\rightarrow \mathcal{F}_t = G_t \vee H_t$

Bigger assumptions:
 D_1 : There exists a G -measurable progressive non-negative process λ such that:
$$P(T_d > t | G_t) = e^{-\int_0^t \lambda(s) ds} \quad \forall t \geq 0$$

Comment: Notice that $P(T_d > t | G_t) < 1$ with this definition meaning that if we only know G_t we don't know if default has happened yet.
But of course we have: $P(T_d > t | \mathcal{F}_t) = P(T_d > t | H_t) = 1_{\{T_d > t\}}$.
 D_2 : $P(T_d > t | G_\infty) = P(T_d > t | G_t)$
Comment: This way we ensure that the G -filtration never learns anything about whether or not default has happened at any given time.

Using D_1 and D_2 we instead define default time as:
$$T_d = \inf \{t \geq 0 \mid \int_0^t \lambda(s) ds \geq \phi\}, \quad \phi \sim \text{Exp}(1)$$

 $\uparrow$
random number drawn from exp distribution

Lemma 12.2:
For any random variable Y with finite expectation, If assumption $D1$ holds we have:
$$E[1_{\{T_d > t\}} Y | \mathcal{F}_t] = 1_{\{T_d > t\}} e^{-\int_0^t \lambda(s) ds} E[1_{\{T_d > t\}} Y | G_t]$$

This allows us to move from $\mathcal{F}_t$ to just G_t , which just models pure economic variables.

- Summarizing Lemmas:
- SL1: Let X be a G -measurable random variable (and r is progressive).
$$E[e^{-\int_0^t r(s) ds} X 1_{\{T_d > t\}} | \mathcal{F}_t] = 1_{\{T_d > t\}} E[e^{-\int_0^t r(s) + \lambda(s) ds} X | G_t]$$

Useful for pricing credit risk payments with Fixed timing.
- SL2: $f_{\omega}(x)$ is the density function for the random variable $U = T_d | G_t \vee H_t$
$$f_{\omega}(x) = 1_{\{T_d > x\}} \lambda(x) e^{-\int_0^x \lambda(s) ds} 1_{\{x \in \mathcal{S}\}}$$

Demonstrates that we can get rid of T_d in many calculation by "integrating" it out.
- SL3: Let $\{Y(t)\}_{t \geq 0}$ be a G -progressive stochastic process
$$E[e^{-\int_0^T r(s) ds} Y(T) 1_{\{T_d \leq T\}}] = 1_{\{T_d > 0\}} \int_0^T E[e^{-\int_0^t r(s) + \lambda(s) ds} Y(t) \lambda(t) | \mathcal{F}_t] dt$$

Used for pricing payoffs that happen at the default time T_d .

Zero-Recovery:
Let X be a payment at time T . Zero recovery refers to a situation where you get nothing if $T_d \leq T$ (Say Counterparty defaults).
Value of X without credit risk: $X(t,T) = E_t^Q[e^{-\int_t^T r(s) ds} X_T | \mathcal{F}_t]$
Use SL1 to account for credit risk: $C_d(t,T) = E_t^Q[e^{-\int_t^T r(s) + \lambda(s) ds} X_T 1_{\{T_d > T\}} | \mathcal{F}_t]$
$$= 1_{\{T_d > t\}} E_t^Q[e^{-\int_t^T r(s) + \lambda(s) ds} X_T | \mathcal{F}_t]$$

Adding default risk here just amounts to discounting with $r + \lambda$.
Partial Recovery at Maturity:
Let X be a payment at time T . Let $R(t) \in (0,1)$ be a stochastic process. If default happens at $t = T_d \leq T$ we will get $XR(T)$ at time T . $R(t)$ essentially measures the percent recovery in the event of default.
Price:
$$C(t,T) = E_t^Q[e^{-\int_t^T r(s) ds} X (1_{\{T_d > T\}} + R(T) 1_{\{T_d \leq T\}}) | \mathcal{F}_t]$$

$$= C_0(t,T) + E_t^Q[e^{-\int_t^T r(s) ds} XR(T) 1_{\{T_d \leq T\}} | \mathcal{F}_t]$$

Let $R(t,T) = E_t^Q[R(T) | G_t]$ and make R independent of r and λ so it can be moved outside the expectation and split up the indicator:
$$C(t,T) = C_0(t,T) + R(t,T) E_t^Q[e^{-\int_t^T r(s) ds} X (1_{\{T_d > T\}} - 1_{\{T_d > T\}}) | \mathcal{F}_t]$$

$$= C_0(t,T) + R(t,T) (X_0(t) 1_{\{T_d > T\}} - C_0(t,T))$$

Partial Recovery at Default:
Same as before but Recovery happens at default.
$$C_d(t,T) = E_t^Q[e^{-\int_t^T r(s) ds} X(T) 1_{\{T_d > T\}} + e^{-\int_t^{T_d} r(s) ds} X(T_d) R(T_d) 1_{\{T_d \leq T\}} | \mathcal{F}_t]$$

Use SL3:
$$C_d(t,T) = C_0(t,T) + 1_{\{T_d > t\}} \int_t^T E_t^Q[e^{-\int_t^u r(s) + \lambda(s) ds} X(u) \lambda(u) | \mathcal{F}_t] du$$

$$= C_0(t,T) + C_D(t,T)$$

Affine processes:
Recall the general transformation function for an affine Process Y :
$$f(Y(t), t; T) = E[e^{-\int_t^T (p_0 + p_1 Y(t) + w_1 + w_2 Y(t)) ds} | \mathcal{F}_t] = e^{A(t,T) + B(t,T) \cdot Y(t)}$$

Where A and B solve the Riccati ODEs. We can construct the following model:
$$r(t) = \bar{r}_0 + \bar{p}_1 \cdot Y(t), \quad \lambda(t) = \pi_0 + \pi_1 \cdot Y(t)$$

Now if we identify:
 $p_0 = \bar{r}_0 + \pi_0, \quad p_1 = \bar{p}_1 + \pi_1, \quad w_0 = 0, \quad w_1 = 0$
We get the transform we need:
$$f(Y(t), t; T) = E[e^{-\int_t^T (r(s) + \lambda(s)) ds} | \mathcal{F}_t] = e^{A(t,T) + B(t,T) \cdot Y(t)}$$

But what about expressions like $E[e^{-\int_t^T r(s) + \lambda(s) ds} \lambda(t) | \mathcal{F}_t]$?
 $\uparrow$
annoying term
 $\cdot$ Fix 1: $E[e^{-\int_t^T r(s) + \lambda(s) ds} + \lambda(t) | \mathcal{F}_t]$
 $\cdot$ Fix 2 (better): $E[e^{-\int_t^T r(s) + \lambda(s) ds} + \lambda(t) | \mathcal{F}_t]$
Lemma 12.10
$$E[e^{-\int_t^T r(s) + \lambda(s) ds} \lambda(t) | \mathcal{F}_t] = (\partial_r A(t, T, 0) + \partial_r B(t, T) \cdot Y(t)) e^{A(t, T, 0) + B(t, T) \cdot Y(t)}$$

Partial Recovery at Default revisited

If we assume the cash flow and recovery all are exponential functions of the state vector Y in the following was
$$X(t) := e^{h_0 + h_1 \cdot Y(t)}, \quad R(t) := e^{h_2 + h_3 \cdot Y(t)}$$

We can set now
$$\rho_0 = \bar{r}_0 + \pi_0, \quad \rho_1 = \bar{p}_1 + \pi_1, \quad w_0 = h_0 + j_0 + X \pi_0, \quad w_1 = X \pi_1 + h_1 + j_1$$

$$\mathbb{E}^Q[e^{-\int_t^T (r(s) + \lambda(s)) ds} + X(t) R(t) | \mathcal{F}_t] = \mathbb{E}^Q[e^{-\int_t^T (\rho_0 + \rho_1 \cdot Y(s) + w_0 + w_1 \cdot Y(s)) ds} | \mathcal{F}_t] = e^{A(t,T,u,v) + B(t,T,u,v) \cdot Y(t)}$$

And following the same steps as on the previous slide (29) becomes
$$C_D(t,T) = 1_{\{T_d > t\}} \int_t^T \mathbb{E}^Q[e^{-\int_t^u (r(s) + \lambda(s)) ds} X(u) R(u) \lambda(u) | \mathcal{F}_t] du$$

$$= 1_{\{T_d > t\}} \int_t^T \frac{\partial}{\partial x} \mathbb{E}^Q[e^{-\int_t^u (r(s) + \lambda(s)) ds} + X(u) R(u) | \mathcal{F}_t] \Big|_{x=0} du$$

$$= 1_{\{T_d > t\}} \int_t^T (\partial_r A(t, u, 0) + \partial_r B(t, u, 0) \cdot Y(t)) e^{A(t,u,0) + B(t,u,0) \cdot Y(t)} du$$

The $A(t, u, 0)$ and $B(t, u, 0)$ are given from the Ricatti ODE's and $\partial_r A(t, u, 0)$ and $\partial_r B(t, u, 0)$ can be calculated numerically.

Credit default swaps:
A CDS is a contract where the buyer pays a fixed coupon C on a series of dates $T_1, \dots, T_N = T$. This payment stream stops if the underlying bond defaults. This is known as the "premium leg" of the contract. The "protection leg" of the contract kicks in if the bond defaults, paying $1 - R(T_d)$ to the buyer of the contract.

Start by considering the premium leg.
First we have the present value of all future payments:
$$I_1(t) = E[e^{-\int_t^T \sum_{i=1}^N \delta_i \mathbb{1}_{\{T_d > T_i\}} | \mathcal{F}_t}] \quad \delta_i = T_i - T_{i-1}$$

 $\uparrow$
 C is a yearly amount so we normalise by how much time has actually passed.

Second we have a partial payment if the bond issuer defaults in the middle of a coupon payment period ($T_{i-1} < T_d \leq T_i$)
$$I_2(t) = E[e^{-\int_t^T \sum_{i=1}^N r(s) ds} (T_d - T_{i-1}) 1_{\{T_{i-1} < T_d \leq T_i\}} | \mathcal{F}_t]$$

Only one term is non-zero:
$$I_2(t) = E[e^{-\int_t^{T_d} r(s) ds} (T_d - T_d(t-1)) 1_{\{T_d > T_d\}} | \mathcal{F}_t]$$

Where $\eta(t)$ satisfies $T_{\eta(t)-1} \leq t \leq T_{\eta(t)}$ i.e. it is the index of the T after the default happens.

Total value of premium leg:
$$V_{\text{prem}}(t) = C(I_1(t) + I_2(t))$$

Total value of protection leg:
$$V_{\text{prot}}(t) = E[e^{-\int_t^T r(s) ds} (1 - R(T_d)) 1_{\{T_d \leq T\}} | \mathcal{F}_t]$$

Total value of CDS:
$$V_{\text{CDS}} = V_{\text{prot}} - V_{\text{prem}} = V_{\text{prot}} - C(I_1 + I_2)$$

The C -value that makes $V_{\text{CDS}} = 0$ is known as the CDS-"spread":
$$CDS = C = \frac{V_{\text{prot}}}{I_1 + I_2}$$

Strategy to solve system: Use summary Lemmas on I_1, I_2 and V_{prot} , assume affine λ and r and solve.

The CDS-spread is observable and gives us an idea of the probability of default. To see why, consider the simplification:
Let $I_2 = 0$, $\lambda(t)$ deterministic, R constant.
$$V_{\text{prem}} = C \sum_{i=1}^N \delta_i P(T_0, T_i) Q(T_d > T_i)$$

 $\uparrow$
risk-neutral probability that company has defaulted: $e^{-\int_0^{T_i} \lambda(s) ds}$

$$V_{\text{prot}} \approx (1-R) \sum_{i=1}^N \delta_i P(T_0, T_i) Q(T_d > T_i)$$
$$C \approx \frac{(1-R) \sum_{i=1}^N \delta_i P(T_0, T_i) Q(T_d > T_i) \lambda(T_i)}{\sum_{i=1}^N \delta_i P(T_0, T_i) Q(T_d > T_i)}$$
$$\approx \sum_{i=1}^N \lambda_i \lambda(T_i) (1-R)$$

The CDS spread is roughly equal to the loss given default $(1-R)$ multiplied by the weighted average of future default intensities. If we furthermore assume constant intensity λ , the ZCB's cancel and we find:

$$\lambda = \frac{C}{1-R}$$

We can then observe C and estimate R and find the Q -probability that a company survives 5 years:

$$Q(T_d > 5 \text{ years}) = e^{-\int_0^5 \lambda ds}$$

Therefore High CDS-spread $\rightarrow$ Higher probability of default.

LIBOR:

The London Intra-bank Offering Rate was the estimated average rate that large banks in the UK could borrow at. This was used as a proxy for the forward rate $f(t, \text{new}(T))$. Banks however could be close to defaulting or have bad liquidity, exposing LIBOR to credit and liquidity risk. It is now discontinued and we use overnight rates instead, which are based on transactions that happened overnight.

Example on LIBOR:
One could create a contract where we exchange 3m LIBOR rates with 6 month LIBOR rates. If LIBOR rates were risk-free, the spread should be 0, which they essentially were until 2008. Suddenly the 6-month rates reflected the large credit risks of major banks.
One fix is to move to a multi-curve model $\rightarrow$ Model short rate $r(t)$ AND LIB(t)
For Overnight-indexed swaps we can just calibrate $r(t)$ separately.
For Interest-rate swaps we price using a joint model for $r(t)$ and $L(t, T)$.

Central Counterparties (CCP)
80% of interest rate swaps are traded through a middleman (CCP). The CCP holds all the collateral from issuer B if B defaults on their contract with A. Collateral here is what B should pay to A if contract is terminated today. A is allowed to reinvest the collateral, while B is paid a growth rate on their collateral corresponding to the collateral rate. Filipovic and Trolle show that pricing swaps in this model essentially just amounts to discounting with the collateral rate:

$$V(t) = E[e^{-\int_t^T r(s) ds} X | \mathcal{F}_t]$$

$\uparrow$
Value of derivative from point of receiver

Risk-neutral OIS-rate:
$$OIS(t,T) = \frac{\sum_{i=1}^N E[e^{-\int_t^{T_i} r(s) ds} \delta_i 1_{\{T_d > T_i\}}]}{\sum_{i=1}^N E[e^{-\int_t^{T_i} r(s) ds} \delta_i]}$$

 $\uparrow$
depends on r

This is the fair fixed swap rate for an OIS contract that pays $\delta_i (L(T_{i-1}, T_i) - OIS(t, T_i))$ at each payment term. Note that all terms here depend on the collateral rate $r(t)$. For IRS contracts we need a multicurve model. One for $L(t, T)$ and another for $r(t)$. The reason is that $L(t, T)$ contains risks previously discussed.

So we want:
$$\underbrace{\text{Libor}}_{\text{IRS}} = \underbrace{\delta}_{\text{OIS}} (\underbrace{\text{Interest rate risk}}_{\text{OIS}} + \underbrace{\text{credit risk}}_{\text{CDS}} + \underbrace{\text{Liquidity risk}}_{\text{Residual}})$$

Filipovic and Trolle model the risk-free rates as:
$$dr(t) = K_r (y(t) - r(t)) dt + \sigma_r dW_r(t)$$

$$dy(t) = K_y (\theta_y - y(t)) dt + \sigma_y (\rho dy_r(t) + \sqrt{1-\rho^2} dW_y(t))$$

 $\uparrow$
2-factor Gaussian ATM $\leftarrow$ proved in paper.
The collateral rate is given by $r_c(t) = r(t) + \Lambda$. Λ can be thought of as the base line default intensity $\lambda(t, t_0) = \Lambda$. At $t > t_0$ a member of the LIBOR-panel may experience a downgrade, that leads to them getting kicked out i.e. $\lambda(t, t_0) > \lambda(t_0, t_0) = \Lambda$. That event is relatively rare though and are modelled as a jump process with dynamics:
$$\lambda(t, t_0) = K_\lambda (\Lambda - \lambda(t_0, t_0)) dt + dJ_\lambda(t, t_0)$$

Where the jump process is dictated by the random variables:
$$dV(t) = K_V (\mu_V(t) - V(t)) dt + \sigma_V \sqrt{V(t)} dW_V(t)$$

$$d\lambda(t) = H_0 (\theta_\lambda - \lambda(t)) dt + \sigma_\lambda \sqrt{\lambda(t)} dW_\lambda(t)$$

Combining these equations into a 6-dimensional state vector X_t . Despite being massive and containing jumps this system can be solved as an affine model by solving Riccati ODEs. They get a final LIBOR which allows them to figure out what kinds of risks are driving rates. They model the LIBOR rate like:

$$L(t, T) = \frac{1}{T - t_0} \left(\frac{1}{B(t, T)} - 1 \right) \equiv (t_0, T)$$

$\uparrow$
to become it results every payment term $\uparrow$
defaultable BCB $\uparrow$
Credit and interest risk $\uparrow$
depends on δ

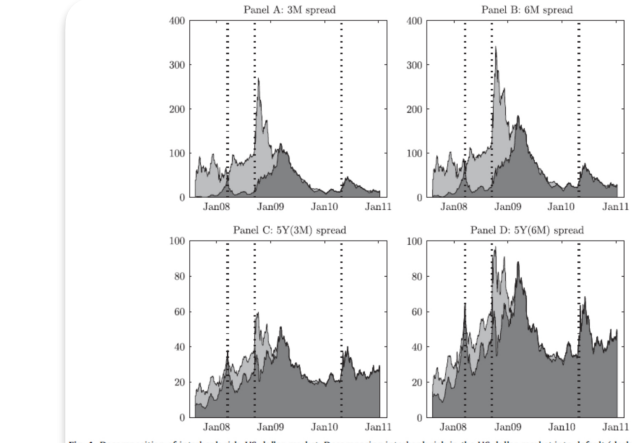


Fig. 4. Histograms of spreads (in bps) for different contracts. The x-axis represents the spread in basis points, and the y-axis represents the frequency. Panel A: LIBOR spread, Panel B: OIS spread, Panel C: IRS spread, Panel D: CDS spread.

$\uparrow$
IRS-OIS spread

The Two Major Discoveries About the GFC
1. The Short-Term Story: A Crisis of Liquidity (Panels A & B)
Liquidity Risk and Liquidity Premium: Banks stopped lending to each other not because they thought the other bank would default in the next 3 months, but because they were terrified they would need that cash themselves to survive. Everyone hoarded cash, driving the liquidity premium through the roof.
2. The Long-Term Story: A Crisis of Credit (Panels C & D)
How bad it really is: C and D, which show how long term lending (5-year rates).
Notice that the dominant color here is completely different. The area is almost entirely dark grey.
What this means: Over a 5-year horizon, short-term cash hoarding (liquidity) doesn't matter as much. If you are lending money for 5 years, your primary concern is whether the bank is still there. The borrowing bank will go bankrupt before the 5 years are up. Therefore, credit risk dominates the long end of the curve.
The Big Exam Takeaway
If you need to summarize this for your professor, list these:
The graph empirically proves that interest rate is term-dependent. During the GFC, the massive spikes in short-term rates (ZIMM) were primarily driven by liquidity risk (cash hoarding), while long-term rates (OIS) were primarily driven by credit risk (fear of actual default). Because of this split, you cannot use a single curve to price derivatives; you need a model that separates and accounts for both.